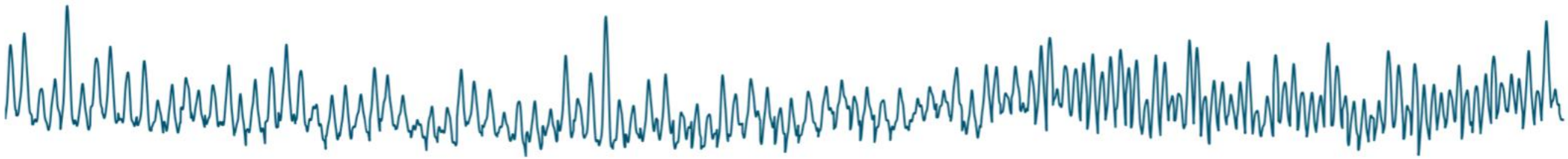
noéa

MEETING OF WATERS

Reading water as rhythm, memory and mood

Affective computing for water systems



Hong Kong, 1884–2025: one beat per year. Height is the year's peak daily rainfall; spacing is the time since the previous peak.

Data can show that a river is changing. It rarely makes anyone feel it.



Noéa works in the missing layer: turning measurement into meaning people can experience.

It reads a record the way a cardiologist reads an ECG

01

A long record

Daily rainfall or river discharge, covering decades to two centuries.

02

ECG-inspired diagnostics

Rainfall: annual peaks, the gaps between them, return periods.

Rivers: flow duration, recession, the seasonal cycle.

03

Two affective values

Valence: pleasant to unpleasant.

Arousal: calm to activated.

Set by design in the curatorial layer.

04

XR experience

Two XR prototypes exist.

Next: connect them to the analysis and to real scenarios.

Valence and arousal follow Russell's circumplex model of affect: Russell, J. A. (1980), Journal of Personality and Social Psychology, 39(6), 1161–1178. Adopted in *Noéa* Issue 4.

Designed to be universal

Each record fits its own model on its own history. Other long rainfall or river records are designed to go through the same pipeline.

Hong Kong rainfall	Public record, Hong Kong Observatory. Daily, 1884–2025.
Rhine and alpine rivers	Public records, GRDC and FOEN. Daily, 8 stations.
The next record	TBA

Two layers, kept deliberately apart

Diagnostic layer

SCIENCE

Rainfall, Hong Kong 1884–2025: annual peaks, the gaps between them, return periods.

Rivers, Rhine and alpine, 8 stations: flow duration, recession, the seasonal cycle.

Measures what happened, and is never changed to suit the design.

Curatorial layer

DESIGNED

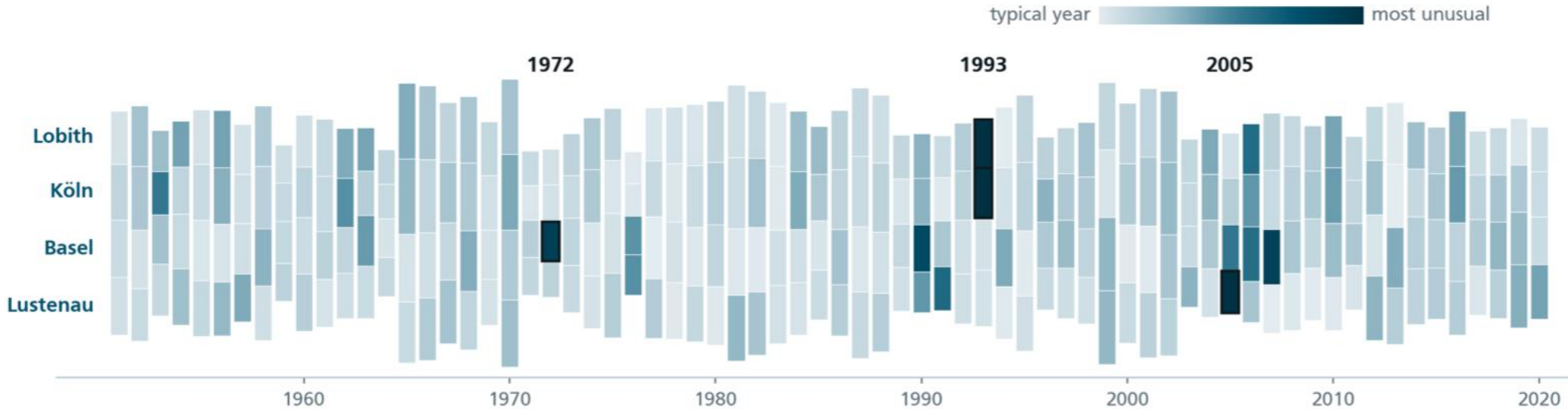
Decides how a result should feel: how alarming, how heavy.

Developed separately in Unity3D, for presentation and exhibition.

Output: valence and arousal. Set by design judgement and checked against named historical events; not yet tested with viewers.

Machine learning finds unusual years

Unsupervised: learn what a typical year looks like for each record, then score how badly each year fits.



Outlined: each station's highest-scoring year, 1951–2020. The model was never told what happened in any of them.

Method: PCA reconstruction error, one model per station. 1993: Christmas Rhine flood (Köln, Lobith). 2005: Alpine floods (Lustenau). 1972: a drought year in Issue 6.
Limit: it flags unusual combinations, not extremes. 1976 drought: 3rd of 70 years at Lustenau, 187th of 208 at Köln. 1995 Köln flood: 101st.

The end result is how people respond

Noéa links machine intelligence to natural intelligence.

Do people feel what it flags? That is the research.

